

## Data Collection and Preprocessing Phase

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|---------------|--------------------------------------------------------------------------------------|
| Date          | 07 Oct 2025                                                                          |
| Team ID       | Xxxxxx                                                                               |
| Project Title | Predicting Plant Growth Stages with Environmental and Management Data Using Power BI |
| Maximum Marks | 2 Marks                                                                              |

## Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

### Data Collection Plan Template

| Section              | Description                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Project Overview     | The <b>Plant Growth Data Classification</b> project aims to analyze and classify plant growth stages based on environmental and soil parameters such as temperature, humidity, sunlight exposure, and soil nutrients. The objective is to build a data-driven model that helps in predicting plant growth performance and identifying the key environmental factors affecting it. |
| Data Collection Plan | The dataset is collected from <b>publicly available agricultural data repositories</b> , with the main source being <b>Kaggle's Plant Growth Data Classification Dataset</b> . This dataset compiles plant growth information recorded under various environmental conditions. Supplementary reference data may be used                                                           |

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|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                             | from online agricultural research repositories and open-source environment datasets for validation and enhancement of model accuracy.                                                                                                                                                                                                                                                                  |
| Raw Data Sources Identified | The primary dataset used is the <b>Plant Growth Data Classification</b> dataset, which contains multiple environmental variables. The dataset is available in CSV format and can be accessed publicly from <b>Kaggle</b> . It includes features such as sunlight hours, temperature, humidity, soil nutrients, and plant growth stage labels, which will be used for preprocessing and model building. |

### Raw Data Sources Template

| Source Name | Description                                                                                                                                                                                                                                                              | Location/URL                                                                                                                                                                  | Format | Size   | Access Permissions |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------------------|
| Dataset 1   | The <b>Plant Growth Data Classification Dataset</b> includes detailed plant growth information with attributes such as temperature, humidity, sunlight hours, soil quality, and growth stages. It is used for data analysis, preprocessing, and classification modeling. | <a href="https://www.kaggle.com/datasets/gorororororo23/plant-growth-data-classification">https://www.kaggle.com/datasets/gorororororo23/plant-growth-data-classification</a> | CSV    | ~50 GB | Public             |

|           |                                                                                                                                                                  |                                                                                                                                                                 |     |        |        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|--------|
| Dataset 2 | Supplementary<br><b>Environmental Reference Data</b> for temperature and humidity variations, used for validation and comparative analysis during preprocessing. | <a href="https://www.kaggle.com/datasets/sobhanmoosavi/us-weather-events">https://www.kaggle.com/datasets/sobhanmoosavi/us-weather-events</a>                   | CSV | ~1 GB  | Public |
| Dataset 3 | <b>Agricultural Soil Nutrient Dataset</b> providing reference nutrient values (NPK) across various soil types to support soil-based plant growth predictions.    | <a href="https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset">https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset</a> | CSV | ~35 GB | Public |